

Conference Paper Title*

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Konstantina Koulaktsidou

Department of Electrical and Computer Engineering
National Technical University of Athens
email

Fragiska Kiminou

Department of Electrical and Computer Engineering
National Technical University of Athens
email

Nikolaos Moraitis

Department of Electrical and Computer Engineering
National Technical University of Athens
email

Abstract—Graph embedding techniques are important tools for representing graphs in low-dimensional vector spaces. This study evaluates three methods, Graph2Vec, NetLSD and Graph Isomorphism Networks (GIN), on benchmark datasets IMDB-Multi, ENZYMES and MUTAG. Performance is assessed through graph classification and clustering tasks, considering predictive accuracy, computational efficiency and scalability. Results indicate that GIN achieves ??? the highest ??? classification accuracy but requires substantial computational resources. Graph2Vec balances performance and efficiency, while NetLSD provides competitive clustering performance with minimal overhead. Therefore it is apparent that each method has its distinct trade-offs. All methods maintain?? reasonable robustness under moderate perturbations, with sensitivity depending on dataset characteristics. These findings provide practical insights that aid in selecting graph embedding approaches based on the task requirements, dataset properties and computational constraints.

Index Terms—Graph Embedding, Graph2Vec, NetLSD, GIN, Graph Classification, Graph Clustering

I. INTRODUCTION

Write your introduction here.

II. PREPARATION

Your main content goes here.

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